LLMs

# AI Engineer

# Interview Questions & Answers



DEVENDHIRAN

LLMs

# What is an LLM?

LLM stands for Large Language Model.

It is an AI model trained on huge amounts of text data to understand and generate human-like language.

## LLMs can:

- Answer questions
- Write code
- Summarize text
- Translate languages
- Build chatbots
- Generate content

## Popular LLMs:

- OpenAI GPT models
- Meta Llama
- Google Gemini
- Anthropic Claude

DEVENDHIRAN

LLMs

# How does an LLM work?

LLMs work using a deep learning architecture called a Transformer.

## Main process:

1. Convert words into tokens
2. Understand relationships between words
3. Predict next token
4. Generate meaningful response

The model learns patterns during training.

## Important Concepts:

- Tokens
- Embeddings
- Attention mechanism
- Transformers
- Probability prediction

DEVENDHIRAN

LLMs

# What is a Transformer Architecture?

Transformer is the core architecture behind modern LLMs.

Introduced in:
"Attention Is All You Need" research paper.

Transformers process words in parallel instead of sequentially.

Main components:

- Encoder
- Decoder
- Self-attention
- Positional encoding

## Why Transformers are powerful?

- Handle long text
- Understand context better
- Train faster
- Scale efficiently

LLMs

# What is Tokenization?

Tokenization means splitting text into smaller pieces called tokens.

## Tokens can be:

- Words
- Subwords
- Characters

## Why Important?

- LLMs do not directly understand text.
- They understand numbers generated from tokens.

## Real-world Example:

- GPT models charge based on token usage.
- More tokens = more cost.

DEVENDHIRAN

LLMs

# What is Prompt Engineering?

Prompt Engineering means designing effective prompts to get better outputs from LLMs.

## A good prompt improves:

- Accuracy
- Context understanding
- Output quality

## Types of Prompting:

- Zero-shot
- One-shot
- Few-shot
- Chain-of-thought

## Example:

❌ Weak Prompt: “Write about AI”

✅ Better Prompt:
“Write a professional LinkedIn post about Generative AI trends in 2026 with a strong hook and hashtags.”

DEVENDHIRAN

LLMs

# Trust takes time, but every post, reply, and story makes a *Difference.*

## Follow for daily updates
## @DEVENDHIRAN